Kevin Wei

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EDUCATION

University of Pennsylvania, School of Engineering and Applied Science, Philadelphia, PA *May 2028*

Bachelor of Science in Engineering in Computer Science & Computer Graphics (GPA: 3.80/4.00)

- *Relevant Coursework*: Discrete Math; Automata Theory; Program Design; Algorithms & Data Structures; Computer Systems; Computer Graphics; Computational Linear Algebra; 3D Figure Modeling; 3D Animation
- *Activities*: UPGRADE, Penn Spark, Penn Lion Dancing, UPenn SIGGRAPH, Content Manager @ Penn Review

TECHNICAL SKILLS

Programming Languages: C#, C++, Java, Python, Lua, C, HLSL, GLSL, JavaScript, TypeScript, HTML, CSS, SQL, OCaml

Platforms/Tools: Unity Game Engine, Unreal Engine, Shader Graph, Qt Creator, Blender, ZBrush, Maya, DaVinci Resolve, After Effects, Aseprite, Audacity, SFXR, MuseScore, Wordpress, Figma, React, Node.js TailwindCSS, Django, Git

RELEVANT EXPERIENCE

Co-President | *UPenn Games Research and Development Environment* (*UPGRADE) Club* *May 2025 – Present*

- **Led meetings and showcases**, oversaw team logistics, engineered architecture for new game, UPGRADE Kart
- Hosted **AI & Gaming guest talk** from CTO and COO of **Medal.tv**, the world's largest gaming clipping tool
- Organized socials, events, a 24-hour Halloween Game Jam, and **secured a sponsorship from Tripo AI**
- **Produced workshops** on game dev basics, level design, modeling, shaders, polishing, UI/UX, and more
- Made dev tools, VFX, and camera code for Catanks, a club-wide Wii-tanks-esque game **published on Steam**

Founder & President | *Maria Carrillo High School (MCHS) Game Dev Club* *September 2021 – June 2024*

- **Founded new club** to teach C#, Unity, game design, pixel art, and music production to my local community
- Organized **guest talk by professional VFX artist** from Hidden Leaf Games to educate about games industry

PROJECTS

Raymarched Fluid Renderer | *C#, HLSL, Unity* *June 2025 – August 2025*

- Simulated Newtonian fluids using **Smoothed Particle Hydrodynamics** and Navier-Stokes equations
- Optimized runtime with bitonic merge sort and **3D spatial hashing** to support **260,000+ particles at 200 FPS**
- Real-time raymarching with **realistic refraction/reflection** and Fresnel equations for accurate light simulation

Flocks of Fish | *C#, HLSL, Unity* *May 2025 – June 2025*

- Developed **GPU-accelerated swarm AI** using Boid principles (separation, alignment, cohesion)
- Implemented **parallel processing** within compute shaders for stable performance of **20,000+ fish at 100 FPS**
- Created procedural fish, sharks and whales, caustics, and an interactive submarine to explore the ocean floor

Flying Through Clouds | *C#, HLSL, Unity* *April 2025 – May 2025*

- Soar through **physically realistic clouds**, with raymarching to simulate forward scattering and light attenuation
- Built GPU-accelerated **Worley Noise and fractal Brownian Motion** tool to generate custom 3D cloud textures

A Bear Game | *C#, Shader Graph, VFX Graph, Unity, Blender* *December 2024 – April 2025*

- Play as a bear who takes pictures of nature, islands, volcanoes, birds, fish, cherry blossoms, and skyscrapers
- Explore the **open world**: beaches, cities, forests; drive on roads, talk to people, catch fireflies, and buy donuts
- Developed **node-based dialogue system** to enable easy, modular production of **complex NPC conversations**
- Built Unity path creation tool with cubic **Bézier splines** and mesh snapping to simplify NPC navigation
- Created **stylized toon shading** integrating Lambertian, Blinn-Phong, and subsurface scattering calculations

Descent | *C#, Unity, Team of 4* *August 2024 – November 2024*

- 3D snowboarding game with endless **procedural terrain**, mid-air tricks system, and sand/snow biomes
- Composed **original 6-minute soundtrack** on MuseScore with 2 cellos, piano, harp, acoustic bass, and drum set